# Isaac J. Newton



## Electronic & Computer Engineer


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### PROFILE

Electronic and Computer Engineer (B.Eng., University of Nigeria, Nsukka) working across embedded systems, electronics, backend engineering and full-stack development. I build reliable software and intelligent hardware, design scalable systems, and turn ideas into functional products with real-world impact.

### EXPERIENCE

**Backend Engineer · IEU Africa** — Remote · 2025 – Present

- Build and maintain scalable backend services and APIs for modern web applications.
- Implement authentication systems and secure access to platform features.
- Run and maintain cloud-based infrastructure.

**Software and Hardware Engineer · OneGrid Energies** — Remote · 2024 – Present

- Develop embedded and energy-monitoring systems.
- Build and maintain scalable web services.
- Interface hardware components with software platforms.
- Work on intelligent power and metering solutions.

**Chief Technology Officer · Switch Electric** — Jul 2021 – Present

- Lead technology development across embedded systems and energy solutions.
- Oversee backend infrastructure and product engineering.

**Electronic Engineer · Arone Energy** — 2021 – 2022

- Designed, developed and tested electronic systems and embedded hardware solutions.
- Handled circuit design and microcontroller integration.
- Troubleshot and debugged electronic systems.

### PROJECTS

**Foot Pressure Measurement Device**

Designed and developed a foot pressure measurement device using PCB design tools. The device captures and analyses foot pressure data for applications in healthcare, sports science and gait analysis, providing insights into foot health and performance.

**Tech:** PCB Design, EasyEDA, Embedded Systems, Sensor Integration, Data Analysis

**SoleSense PWA Insole Monitor** — fluxion9.github.io/solesense/app

Built a progressive web app for SoleSense that connects to the foot pressure measurement device over Bluetooth, collects pressure data, and displays real-time monitoring and analysis for insole performance.

**Tech:** PWA Development, Bluetooth, Data Visualization, Sensor Data, Health Tech

### OneGrid Energies

onegridenergies.org

Built the OneGrid Energies website for a Nigerian clean energy company focused on renewable energy access, solar installations, recycling innovation, and community empowerment across underserved communities.

**Tech:** Website Development, Frontend Development, Responsive Design, Clean Energy

### Veriss Payments

payment.veriss.co

Fintech payment infrastructure for secure, seamless digital transactions. Focused on payment processing, transaction management, and scalable financial operations.

**Tech:** Backend Development, Node.js, Express.js, MongoDB, Paystack API

### BoqBox

boqbox.ng

Worked on BoqBox, an e-commerce and gifting platform that enables users to purchase products for themselves or send them as gifts to others, with support for recurring subscription-based deliveries and seamless online shopping experiences.

**Tech:** Backend Development, Node.js, Express.js, Paystack API, MongoDB

### Aegis pest Detection Rover · Embedded System Engineer

eneontechnologies.com/projects/aegis-pest-detection-rover

Aegis autonomously scans a farm plot, using a computer vision algorithm to identify soil and crop pests as it moves. Each detection sends a notification straight to the owner's phone on Telegram, so action can be taken before the damage spreads. Pests are often spotted too late, after the harvest is already at risk. Aegis patrols the plot consistently, replacing occasional manual inspection with regular, reliable coverage.

- Robotics
- Embedded Systems
- Computer Vision

**Tech:** IOT, Embedded System, ProductInnovation, Robotics and Engineering

### SafeGuard LPG

eneontechnologies.com/projects/safeguard-lpg

A gas monitoring and alarm system that detects LPG leaks, smoke and abnormal temperature, raises a distinct alarm for each and learns its environment to reduce false alarms over time.

## SKILLS

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|---|---|
| **Languages** | C · C++ · Python · JavaScript · PHP · HTML · CSS |
| **Backend & web** | Node.js · Express.js · MongoDB · REST APIs · Paystack API · Progressive Web Apps · Responsive Design |
| **Hardware & embedded** | Embedded Systems · Hardware / Circuit Design · PCB Design · EasyEDA · Microcontrollers · Sensor Integration · Bluetooth · Internet of Things (IoT) |

## EDUCATION

### Bachelor of Engineering (B.Eng.), Electronic and Computer Engineering

Sep 2019 – 2024

University of Nigeria, Nsukka · Nsukka, Nigeria

Portfolio, projects and the latest version of this CV: **fluxion9.github.io**